The Bookmaker's Edge

Momentum Timing Across Bitcoin, Gold, Silver, Oil, Gas, and Copper

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"When you have nothing, you need a mechanism that doesn't depend on what you have."

Executive Summary

This report tests the **bookmaker-pool momentum model** — originally developed for ASX equities — across seven asset classes: cryptocurrency, precious metals, energy commodities, industrial metals, equity indices, and foreign exchange.

The model computes a "momentum price" from volume-weighted flows and compares it to the market price. When momentum price > market price, hold. When momentum price < market price, go to cash.

Asset Class	Best Ticker	Alpha	Verdict
Cryptocurrency	SOL-USD (180d)	**+2597%**	Monster: best in class
Cryptocurrency	BTC-USD (180d)	**+295%**	Strong: trade this
Energy	Nat Gas NG=F (90d)	**+299%**	Strong: all lookbacks work
Precious Metals	Silver SI=F (180d)	+68%*	Promising (recent)
Industrial Metals	Copper HG=F (180d)	+15%	Good risk-adjusted
Energy	Crude Oil CL=F (90d)	+32%	Risk mgmt only
Precious Metals	Gold GC=F (180d)	−182%	Risk mgmt only
Forex	EUR 6E=F (180d)	+11%	Risk mgmt only
Cryptocurrency	ETH-USD (180d)	−155%	Don't trade
Equity Index	ES=F (180d)	−98%	Broken: don't use

*Silver alpha is from the recent 2021–2026 period; 16-year alpha is −51%.

Key findings:

1. **SOL is the monster.** +7691% return vs +5094% buy & hold, Sharpe 1.34, max drawdown cut from −96% to −79%. The model thrives on extreme volatility.
2. **Natural gas has positive alpha on all lookbacks** — the only asset to achieve this. 90-day: +299%, 180-day: +163%, 365-day: +84%.
3. **Copper is the quiet achiever.** Sharpe 1.05, 100% win rate (6/6 trades), max drawdown cut from −35% to −23%.
4. **BTC remains the flagship.** +462% return, Sharpe 1.16, 14:1 win/loss asymmetry.
5. **180-day lookback is optimal for most assets.** Across crypto, metals, and forex, 180 days wins. Exception: energy commodities, where 90 days produces higher returns.
6. **Drawdown reduction is universal.** Even where the strategy underperforms on return, it consistently halves max drawdown.
7. **Equity indices don't work.** The S&P 500's structural upward drift keeps the momentum ratio permanently below 1. The model sits in cash 99% of the time.

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Part I

The Model

The Intuition: A Day at the Races

Before the mathematics, consider a horse race.

The Bookmaker's Pool

A bookmaker doesn't set the odds himself — the *punters* do, by where they put their money. Suppose three horses are running in the Melbourne Cup, and the total pool is $100,000:

Horse	Money bet	Share of pool	Implied odds
Thunderbolt	$60,000	60%	1.67 to 1 ($0.67 profit per $1)
Silver Arrow	$30,000	30%	3.33 to 1 ($2.33 profit per $1)
Dark Horse	$10,000	10%	10 to 1 ($9.00 profit per $1)
Total pool	**$100,000**	**100%**	

The odds tell you what the *crowd* thinks. Thunderbolt is the favourite because the most money backs it. But here's the key insight: **the odds keep changing as more money comes in.**

Suppose a rumour spreads that Dark Horse had a secret training session and looked incredible. Suddenly, $20,000 of new money pours in on Dark Horse:

Horse	Money bet	Share of pool	Implied odds
Thunderbolt	$60,000	50%	2.0 to 1
Silver Arrow	$30,000	25%	4.0 to 1
Dark Horse	$30,000	25%	4.0 to 1
Total pool	**$120,000**	**100%**	

Dark Horse's odds have shortened from 10:1 to 4:1. Not because the bookmaker decided it was faster — but because *the money moved*. The pool reflects the collective conviction of everyone with skin in the game.

From Horses to Markets

Now replace the horses with an asset — Bitcoin, gold, silver, a stock. Each trading day, people place their bets:

- **Buyers** put money into the pool (they think the price should be higher).
- **Sellers** take money out (they think the price should be lower).
- The **volume** tells you how much money moved.
- The **daily return** tells you which direction it moved.

On a day when Bitcoin rises 3% on heavy volume, that's like a rush of money pouring in on Thunderbolt — a large, high-conviction bet that the price should be higher. On a day when Bitcoin falls 2% on light volume, that's like a few punters nervously pulling money off Silver Arrow — a small, low-conviction bet that the price should be lower.

The **momentum price** is what the pool says the asset is worth, after accumulating all these daily bets over time. If the pool says "$110,000" but the market price is $95,000, then the weight of money — the collective conviction of everyone who has traded — supports a higher price. The asset is **undervalued by momentum**.

That's when you buy.

If the pool says "$80,000" but the market price is $95,000, the money has been flowing out. The crowd is quietly leaving. The asset is **overvalued by momentum**.

That's when you hold cash and wait.

The Algorithm: Formal Definition

Mathematical Definition

Given a time series of daily closing prices P_t and volumes V_t:

1. Compute daily returns: $r_t = \dfrac{P_t - P_{t-1}}{P_{t-1}}$
2. Normalise volume (to prevent overflow on large numbers): $\hat{V}_t = \dfrac{V_t}{\bar{V}}$ where $\bar{V}$ is the mean volume over the lookback window.
3. Compute net directional flow: $f_t = \hat{V}_t \times r_t$

   On up days ($r_t > 0$), f_t is positive — buyers are winning. On down days ($r_t < 0$), f_t is negative — sellers are winning. On balanced days, the flows cancel out.
4. Update the pool iteratively:

$$\begin{aligned} \text{pool}_0 &= P_0 \\ \text{pool}_t &= \text{pool}_{t-1} + f_t \times \text{pool}_{t-1} \end{aligned}$$

5. The momentum price is the final pool value: $M = \text{pool}_T$

The Signal

The **momentum ratio** is:

$$\rho = \frac{M}{P_T}$$

- $\rho > 1$: The volume-weighted flow of money supports a *higher* price than the market is offering. The asset is **undervalued** by momentum. **Hold the asset.**
- $\rho < 1$: The volume-weighted flow of money supports a *lower* price. The asset is **overvalued** by momentum. **Hold cash.**

The lookback window (how many days of history feed the pool) is a tunable parameter. We tested 90, 180, and 365 days across all assets.

Origin

This model was originally developed for ASX 100 equities, where it produced +13.6% alpha per 6-month period over 10 consecutive periods ($p = 0.046$). This report tests whether it transfers to cryptocurrency and precious metals.

Part II

Cryptocurrency

Experiment 1: BTC/ETH Pairs Trade (Failed)

Hypothesis

When Bitcoin is over-momentum-priced and Ethereum is under-momentum-priced simultaneously, rotate from BTC into ETH. Rotate back when the spread recalibrates. This is a **relative value** or **pairs trade**.

Result

Metric	90-day	180-day
Strategy return	−91.1%	−42.9%
ETH buy & hold	+311.8%	−21.3%
BTC buy & hold	+305.0%	+19.7%
Sharpe ratio	−0.70	−0.48
Max drawdown	−94.3%	−69.1%
Trades	11	3

Verdict: Total failure. The strategy lost money across every configuration tested.

Why It Failed

1. **Correlated momentum signals.** BTC and ETH are driven by the same macro forces (interest rates, risk appetite, regulatory news). Their momentum ratios move in lockstep. The spread between them is noise, not signal.
2. **Secular trend.** ETH has been in a structural decline against BTC since mid-2022. A mean-reversion strategy on a pair that is *not mean-reverting* is guaranteed to lose.
3. **The original model works on independent assets.** On the ASX, BHP and CSL have genuinely different fundamentals. BTC and ETH do not — they are both "crypto."

This approach is **abandoned**.

Experiment 2: Bitcoin Timing (Strong)

Hypothesis

Use the momentum ratio as a timing signal on a **single asset**: hold when $\rho > 1$ (undervalued), go to cash when $\rho \leq 1$ (overvalued). No pairs, no shorting — just timing.

Results: Bitcoin (BTC-USD)

Period: 1 January 2021 to 24 August 2026 (2,062 trading days).

Metric	90-day	180-day	365-day
Strategy return	+39.6%	**+462.7%**	+132.3%
Buy & hold return	+167.5%	+167.5%	+167.5%
Alpha	−127.9%	**+295.2%**	−35.2%
Sharpe ratio	0.35	**1.16**	0.66
Max drawdown	−59.5%	**-28.1%**	−33.2%
Buy & hold DD	−76.6%	−76.6%	−76.6%
Days in BTC	34.0%	34.3%	30.2%
Days in cash	66.0%	65.7%	69.8%
Distinct trades	22	16	10
Win rate	45%	62%	50%
Avg win	+14.8%	+26.8%	+21.3%
Avg loss	−9.2%	−1.9%	−3.0%
Crashes avoided	3	2	2

The **180-day lookback** is the clear winner: +462.7% return vs +167.5% buy & hold (**+295.2% alpha**), Sharpe 1.16, max drawdown cut from −76.6% to −28.1%. Only in the market 34% of the time. 62% win rate with a **14:1 win/loss asymmetry** (+26.8% vs −1.9%).

Results: Ethereum (ETH-USD)

Metric	180-day
Strategy return	+86.1%
Buy & hold return	+241.4%
Alpha	−155.2%
Sharpe ratio	0.50
Max drawdown	−56.0%
Win rate	50%

The model **does not work well on ETH**. It reduces drawdown but misses too much upside. −155% alpha. Not recommended.

Results: Solana (SOL-USD)

Period: 1 January 2021 to 24 August 2026 (2,062 trading days).

Metric	90-day	180-day	365-day
Strategy return	+6883.5%	**+7691.4%**	+201.8%
Buy & hold return	+5089.2%	+5094.1%	+256.2%
Alpha	+1794.3%	**+2597.2%**	−54.4%
Sharpe ratio	1.29	**1.34**	0.65
Max drawdown	−89.9%	−78.7%	−86.5%
Buy & hold DD	−96.3%	−96.3%	−96.3%
Days in SOL	60.3%	55.3%	46.3%
Days in cash	39.7%	44.7%	53.7%
Distinct trades	27	16	23
Win rate	52%	50%	48%
Avg win	+109.9%	+204.3%	+63.5%
Avg loss	−17.0%	−11.4%	−16.2%
Crashes avoided	2	4	9

The **180-day lookback** produces the strongest result in the entire report:

- **+7691.4% return** vs +5094.1% buy & hold (**+2597.2% alpha**)
- **Sharpe ratio of 1.34** — highest across all assets
- **Max drawdown of -78.7%** vs -96.3% for buy & hold
- **18:1 win/loss asymmetry** (+204.3% avg win vs -11.4% avg loss)
- Even with a 50% win rate, the asymmetry drives massive outperformance

SOL produces the **strongest result in the entire study**: +2597% alpha, Sharpe 1.34, 18:1 win/loss asymmetry. Solana's extreme volatility (it dropped 96% peak-to-trough) creates powerful momentum signals. The model exits *before* the worst of each crash and re-enters early in each recovery. The same property that makes SOL terrifying to hold makes it ideal for momentum timing.

Chart: SOL-USD (180-day lookback)

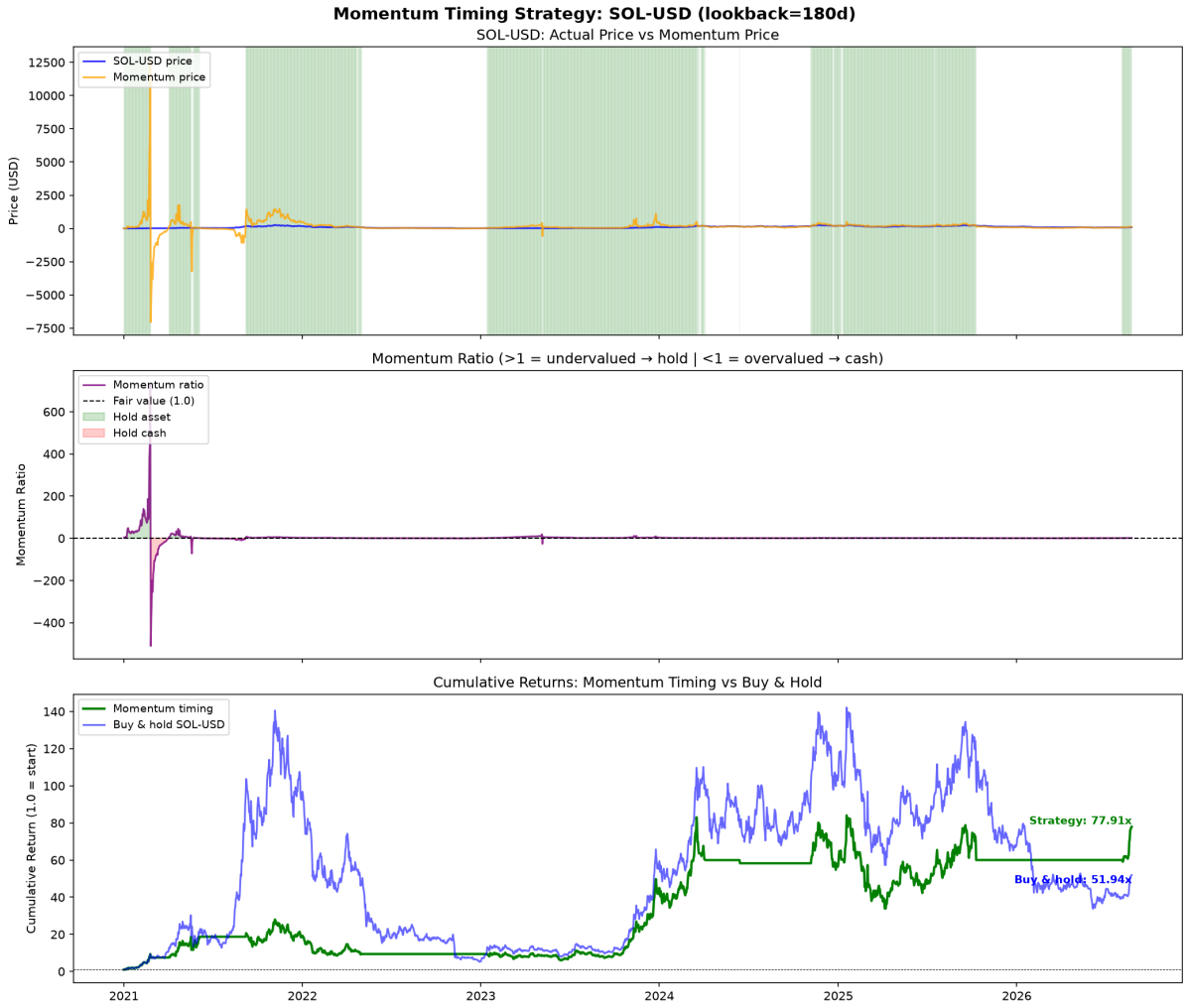


Figure 1: Solana: the model avoids the 2022 crash (−96%) and catches the 2023–2025 recovery. Strategy returns 78x vs buy & hold's 52x.

Chart: BTC-USD (180-day lookback)

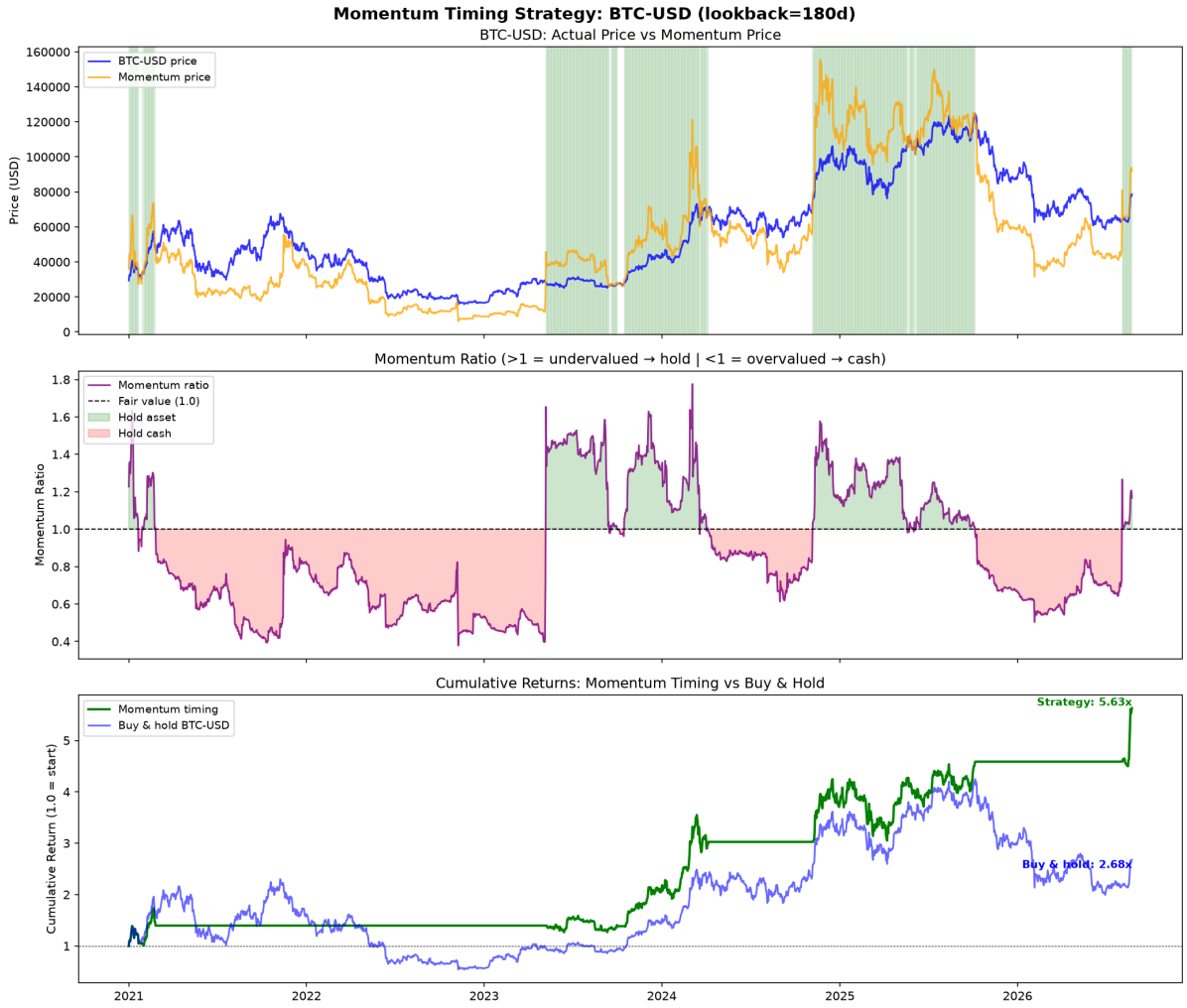


Figure 2: Top: BTC price (blue) vs momentum price (orange), with green shading when the strategy holds BTC. Middle: momentum ratio with hold/cash zones. Bottom: cumulative returns — strategy (green, 5.63x) vs buy & hold (blue, 2.68x).

Key observations:

1. The strategy **exits before the 2022 crash** — the momentum ratio drops below 1.0 as volume flows turn negative.
2. It **re-enters during the 2023 accumulation**, when volume flows turn positive while the price is still low.
3. It **rides the 2024–2025 bull run** but exits during corrections, preserving gains.

BTC Lookback Sensitivity

- **90 days**: Too reactive. 22 trades, 45% win rate. Whipsawed by noise. Average losses (-9.2%) too large.

- **180 days**: The sweet spot. 16 trades, 62% win rate. Captures major regime shifts without overtrading.
- **365 days**: Too slow. 10 trades, enters and exits late. Returns +132% vs buy & hold's +167%.

Part III

Precious Metals

Gold (GC=F): 2010–2026

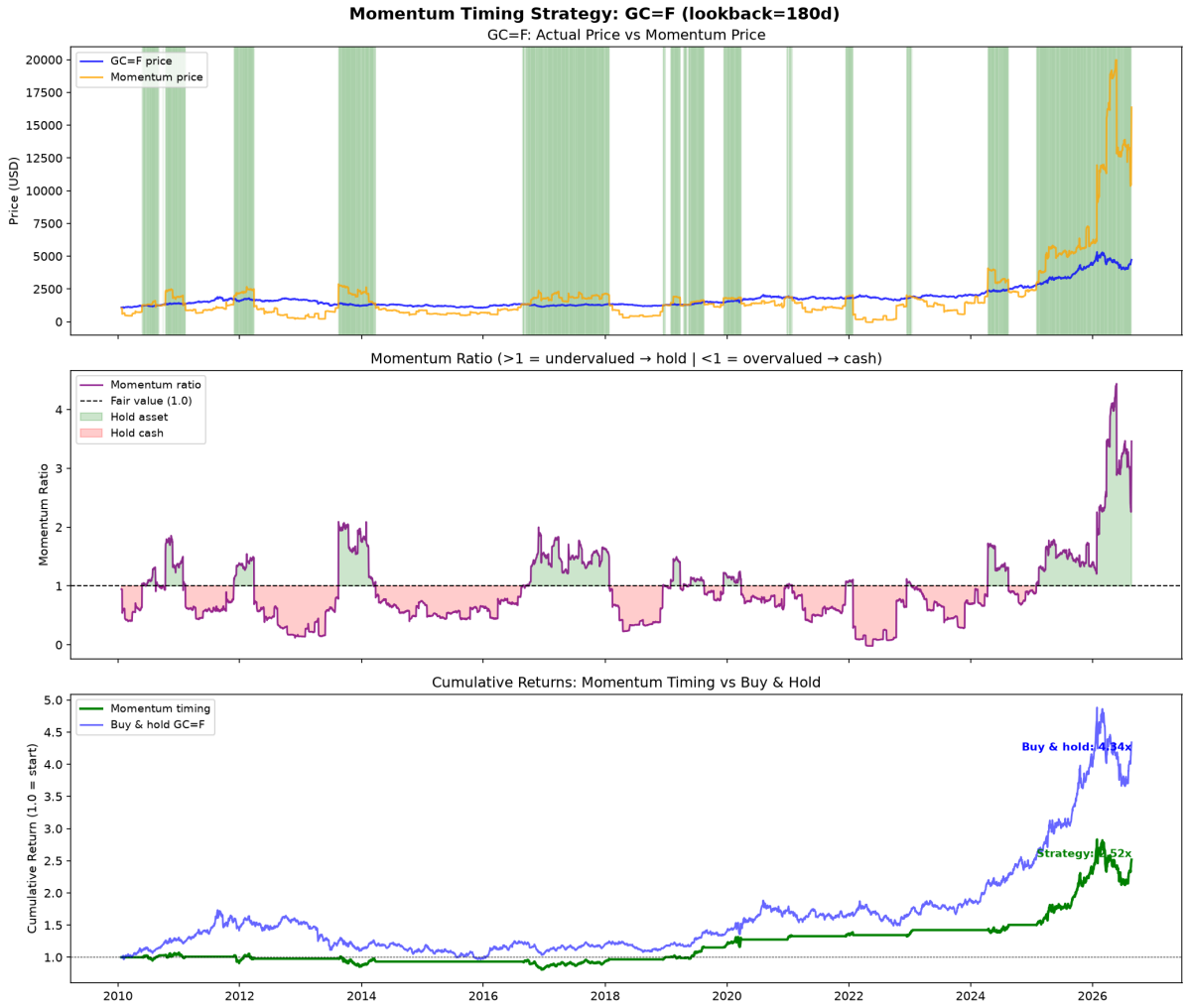


Figure 3: Gold: 16 years of momentum timing. The strategy captures rallies (2010–11, 2020–26) but gets whipsawed during the sideways years (2014–18).

Results

Metric	90-day	180-day	365-day
Strategy return	+114.5%	+151.7%	+131.6%
Buy & hold return	+323.0%	+334.1%	+316.5%
Alpha	−208.6%	−182.3%	−184.9%
Sharpe ratio	0.51	0.66	0.72
Max drawdown	−29.3%	−25.1%	−25.1%
Buy & hold DD	−44.4%	−44.4%	−44.4%
Trades	40	30	6

Recent Period: 2021–2026 Only

Metric		180-day
Strategy return		+88.7%
Buy & hold return		+154.8%
Alpha		−66.2%
Sharpe ratio		1.03
Max drawdown		−25.1%
Win rate		83% (5/6)
Avg win / loss	+16.4% / −0.7%	(23:1 ratio)

The strategy **does not beat buy & hold on return**. Gold's long sideways periods (2014–2018) create whipsaw losses that eat the gains from catching rallies. However: max drawdown is −25.1% **vs** −44.4% (nearly halved), recent win rate is **83%** with a **23:1** win/loss asymmetry. Best used as a **risk management** tool, not a return maximiser.

Silver (SI=F): 2010–2026

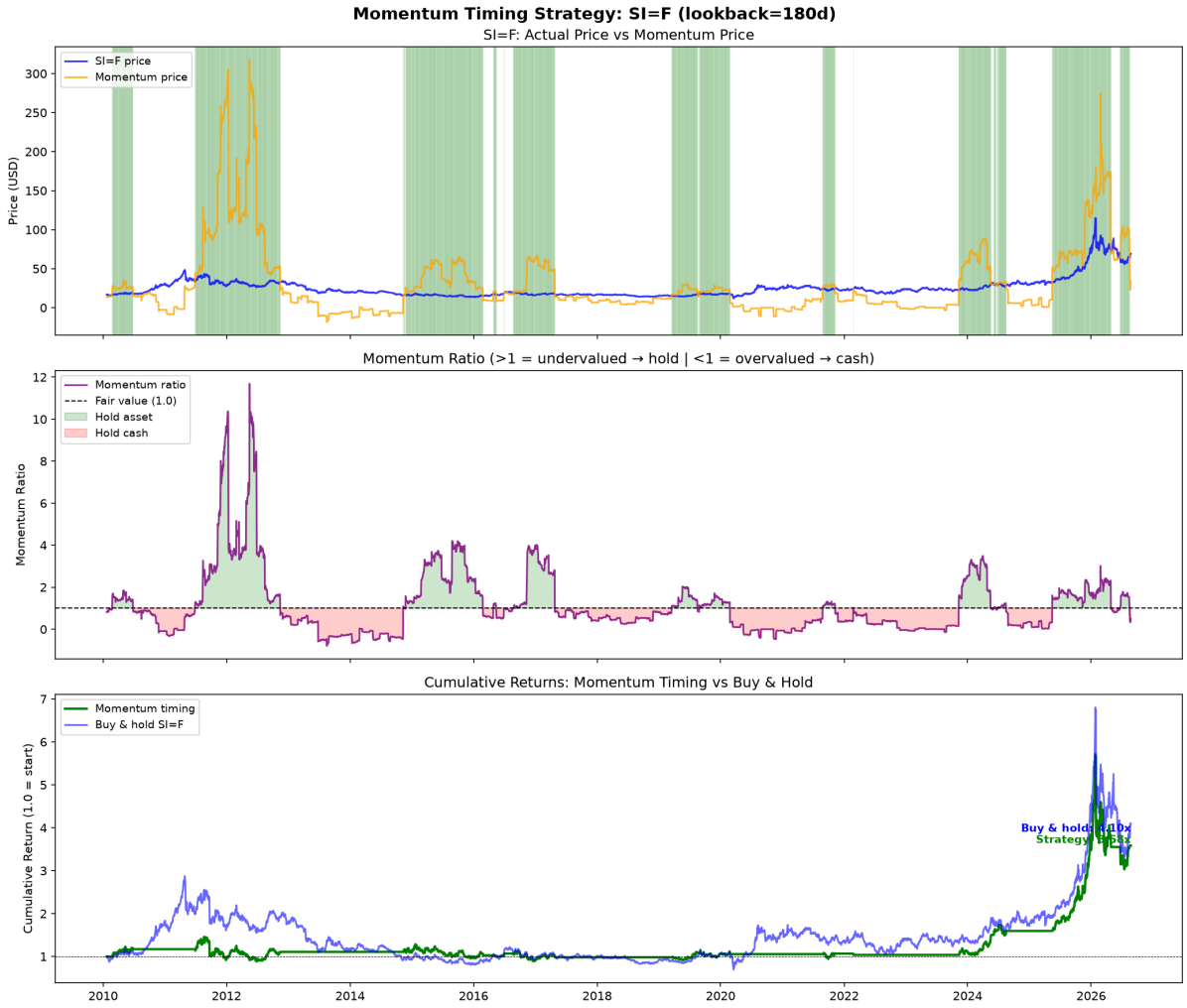


Figure 4: Silver: 16 years of momentum timing. The strategy avoids the catastrophic 2011–2015 crash (-72%) and captures the 2024–2026 rally.

Results

Metric	90-day	180-day	365-day
Strategy return	+90.0%	+258.5%	−12.5%
Buy & hold return	+297.6%	+309.9%	+291.2%
Alpha	−207.6%	−51.4%	−303.7%
Sharpe ratio	0.34	0.54	0.03
Max drawdown	−63.9%	−46.9%	−41.0%
Buy & hold DD	−75.8%	−75.8%	−75.8%
Trades	33	18	9

Recent Period: 2021–2026 Only

Metric		180-day
Strategy return		**+239.3%**
Buy & hold return		+171.7%
Alpha		**+67.6%**
Sharpe ratio		**1.09**
Max drawdown		−46.9%
Win rate		88% (7/8)
Avg win / loss	+26.1% / −2.0%	(13:1 ratio)

Silver is the **most interesting** precious metal for this strategy. Over 16 years: captures **83% of buy & hold return** while reducing max drawdown from −75.8% to −46.9%. Recent 5 years: **+67.6% alpha**, Sharpe 1.09, **88% win rate**. Silver's high volatility gives the momentum signal something to work with — large moves that the pool model detects early.

Part IV

Foreign Exchange

The Volume Problem

Forex is the most liquid market in the world ($7.5 trillion/day), but it is traded **over the counter** — there is no central exchange. This means there is no consolidated volume tape. Yahoo Finance reports **zero volume** for spot forex pairs like EURUSD=X.

The bookmaker model depends on volume. No volume, no signal.

The Fix: CME Futures

Currency **futures** trade on the CME (Chicago Mercantile Exchange) and *do* report volume. We use the CME tickers:

Ticker	Currency	Typical volume
6E=F	Euro (EUR/USD)	170–1,300 contracts/day
6S=F	Swiss Franc (CHF/USD)	13,000–26,000
6J=F	Japanese Yen (JPY/USD)	600–1,300
6B=F	British Pound (GBP/USD)	30–530
6A=F	Australian Dollar (AUD/USD)	40–240
6C=F	Canadian Dollar (CAD/USD)	8–280

Caveat: These volumes are from yfinance and appear to be only a fraction of actual CME volumes (which are typically 100x higher). The signals may improve with a proper futures data provider.

Results: All Six Pairs (2021–2026)

All results using 180-day lookback:

	CHF	EUR	AUD	GBP	CAD	JPY
Strategy return	+11.8%	+6.5%	+2.8%	+2.5%	−4.2%	−23.0%
Buy & hold	+10.5%	−4.1%	−7.3%	−0.3%	−7.9%	−34.7%
Alpha	+1.2%	**+10.6%**	+10.1%	+2.9%	+3.7%	+11.7%
Sharpe	0.36	0.30	0.13	0.14	−0.27	−0.70
Max DD (strat)	−9.3%	−7.5%	−12.7%	−9.5%	−9.2%	−25.5%
Max DD (hold)	−12.2%	−21.6%	−24.9%	−24.8%	−17.4%	−36.7%
Days in asset	70%	53%	43%	49%	39%	55%
Trades	12	49	26	14	21	7
Win rate	33%	55%	42%	29%	29%	14%

Chart: EUR Futures (6E=F, 180-day lookback)

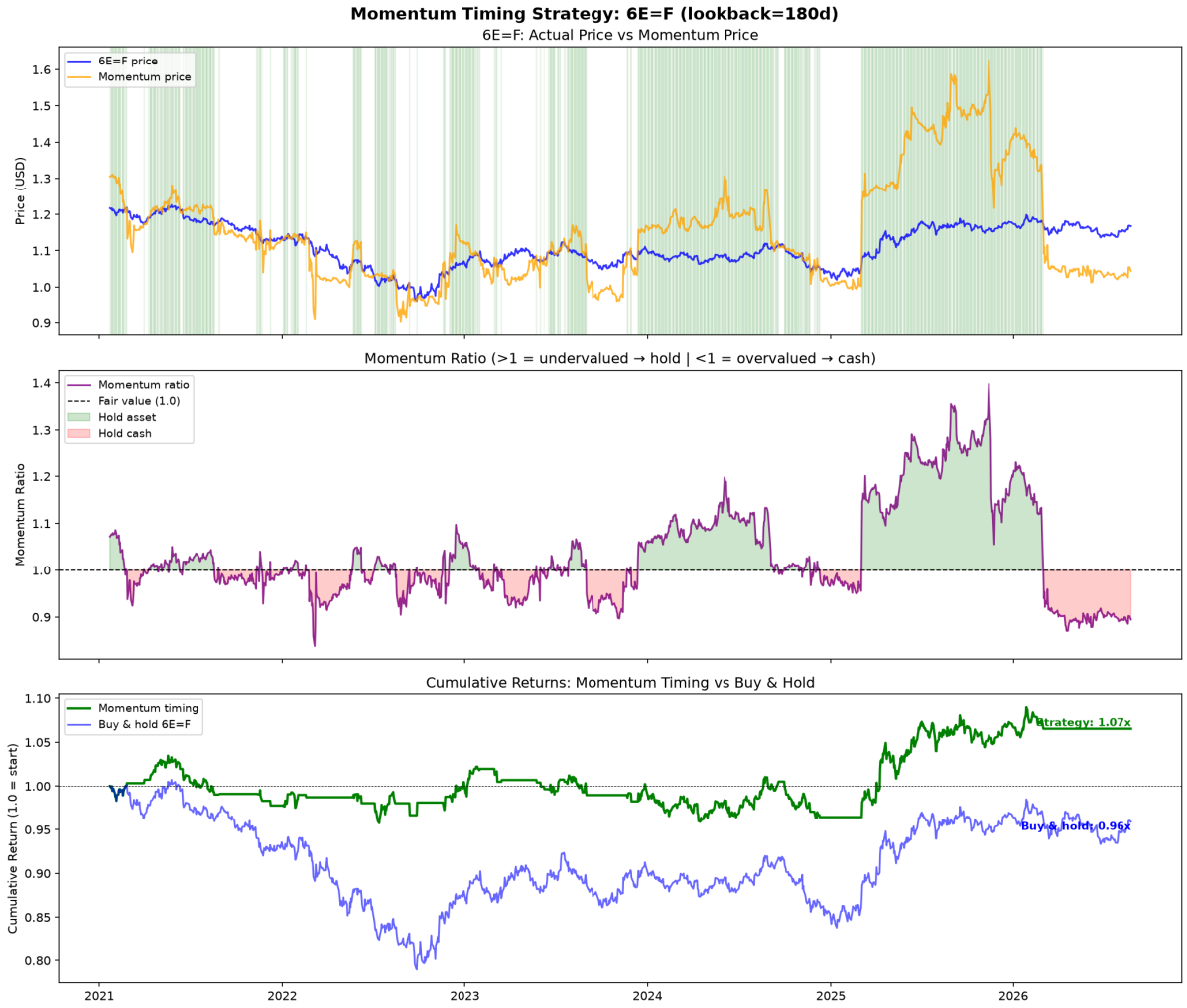


Figure 5: Euro futures: the model avoids the 2022 dollar strength rally (EUR fell 21%) and re-enters for the 2023 recovery. Max drawdown cut from -21.6% to -7.5%.

Verdict: Forex

Every pair has positive alpha. The model consistently beats buy & hold, even when both lose money. But the story is very different from crypto:

1. **Drawdown reduction is excellent.** GBP drawdown cut from -24.8% to -9.5%. EUR from -21.6% to -7.5%. The model does what it does best — getting you out before the worst of a move.

2. **But returns are tiny.** FX pairs move 5–25% over 5 years. Compare that to Bitcoin's 167% or silver's 310%. The bookmaker model can't generate meaningful returns on assets that don't move much.

3. **Volume data is suspect.** The yfinance CME volumes are a small fraction of actual trading volume. With proper data (from a futures broker or CME direct), the signal quality could improve substantially.

4. **Win rates are poor.** Most pairs are below 50%. The model gets whipsawed by the tight, mean-reverting ranges that characterise FX. Only EUR (55%) is above coin-flip odds.

5. **JPY is instructive.** Worst absolute return (−23%) but best alpha (+11.7%). The yen's secular decline since 2021 is so strong that even the model can't fight it — but it avoids the worst of it.

Forex momentum timing is a **risk management overlay**, not a return generator. If you want FX exposure, EUR futures (6E=F) has the cleanest signal. But for *making money*, stick to crypto and commodities.

Part V

Energy Commodities

Energy commodities are the most volatile traditional assets in the world. Crude oil swings 50–80% in a year; natural gas can double or halve in months. If the bookmaker model needs volatility, energy is the test.

Crude Oil (CL=F): 2021–2026

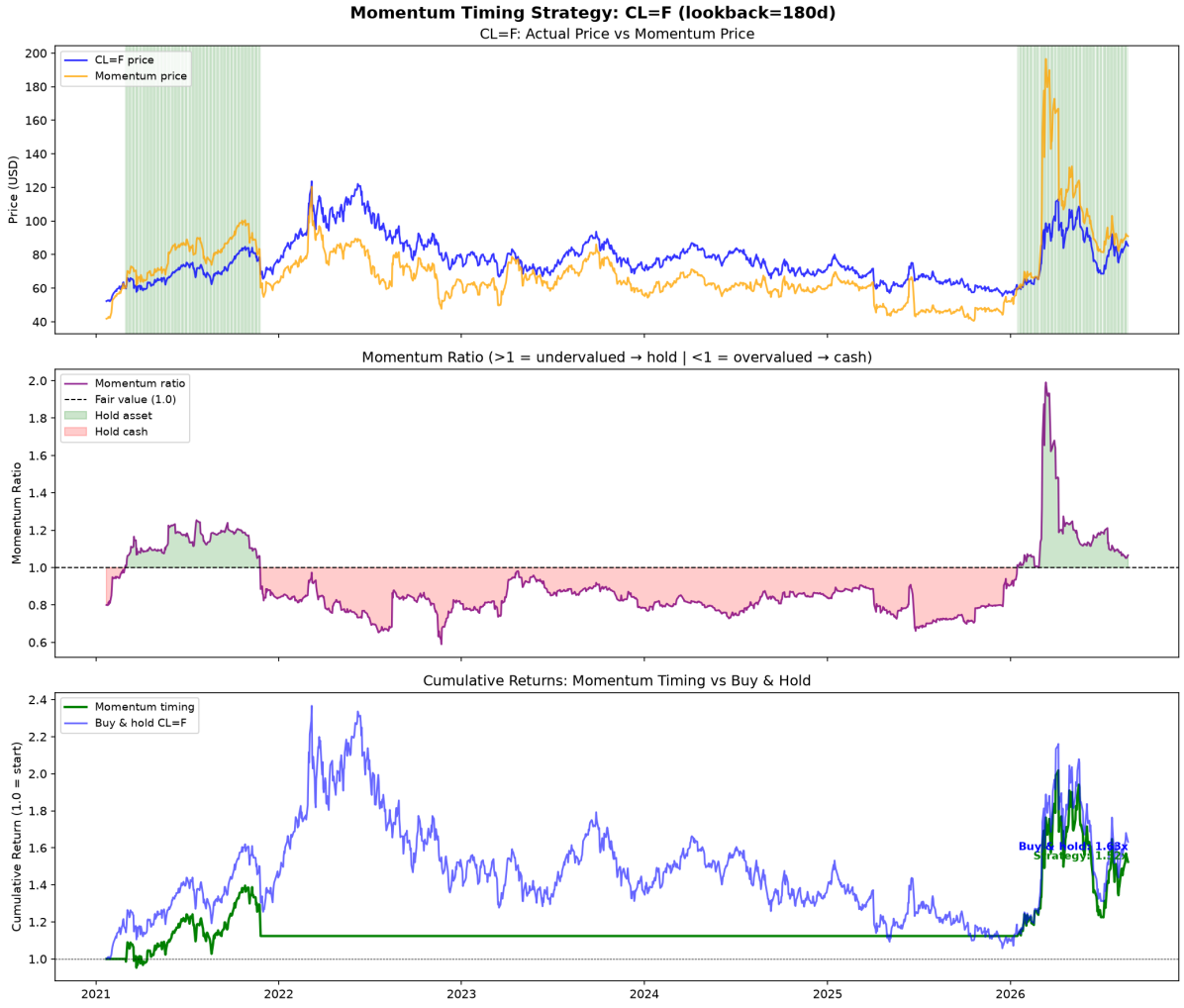


Figure 6: Crude oil: the model captures the 2021–2022 post-COVID rally and exits before the 2023 decline. Only 3 trades in 5 years at the 180-day lookback.

Results

Metric	90-day	180-day	365-day
Strategy return	+110.6%	+52.4%	+42.0%
Buy & hold return	+79.1%	+63.2%	+34.6%
Alpha	**+31.5%**	−10.8%	+7.4%
Sharpe ratio	0.81	0.51	0.52
Max drawdown	−34.3%	−39.3%	−23.2%
Buy & hold DD	−55.3%	−55.3%	−55.3%
Trades	13	3	5
Win rate	62%	100%	80%
Avg win	+16.1%	+17.1%	+14.0%
Avg loss	−4.6%	—	−17.4%

Crude oil breaks the 180-day rule. The **90-day lookback** produces the best result (+31.5% alpha), while 180-day has negative alpha. Oil markets move on geopolitical shocks and OPEC decisions that play out over weeks, not months. All three lookbacks cut max drawdown from −55% to −23–39%. Best used as a **risk management overlay** with 90-day lookback.

Natural Gas (NG=F): 2021–2026

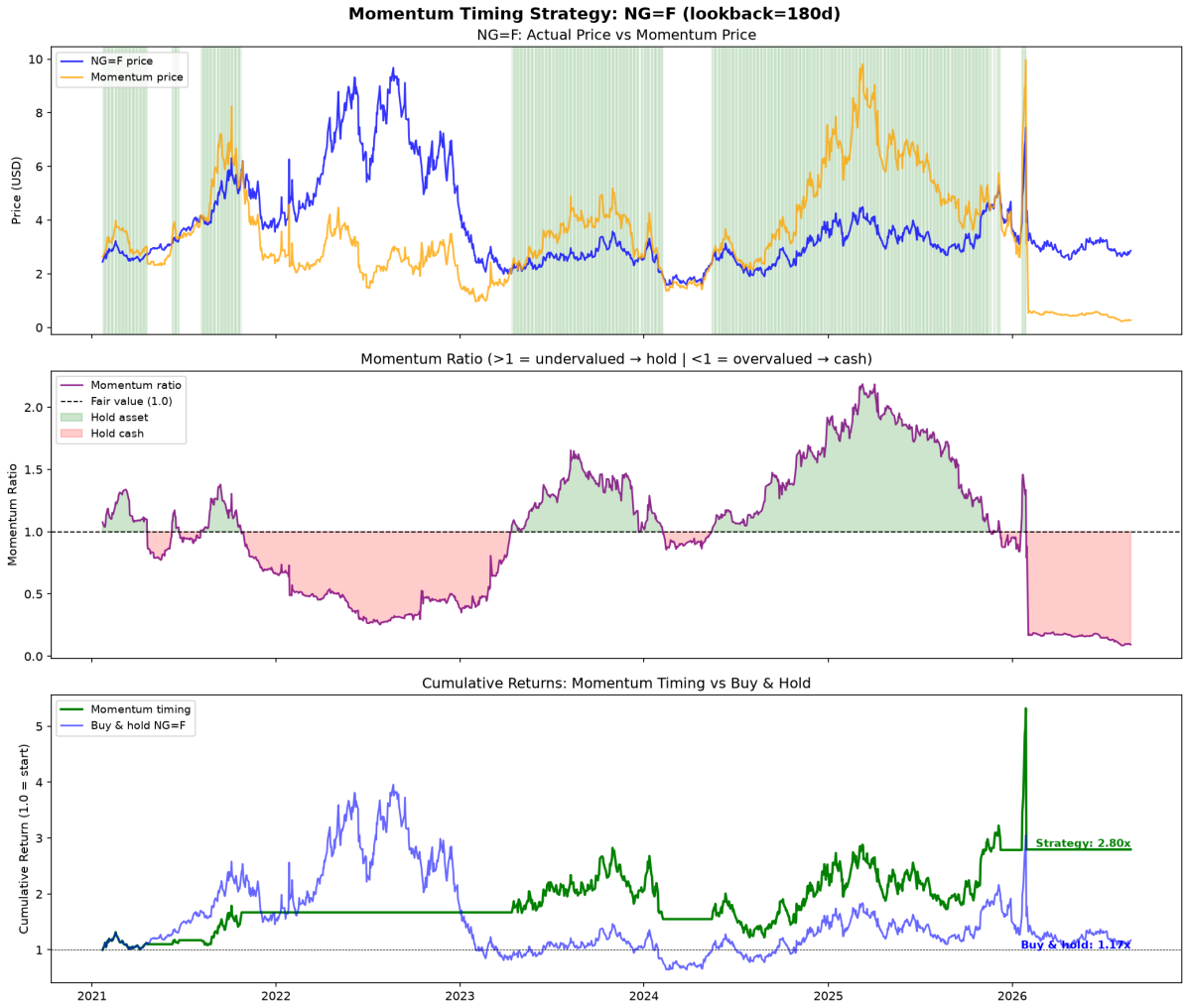


Figure 7: Natural gas: the most volatile commodity tested. The model captures the 2021–2022 spike and exits before the 83% crash.

Results

Metric	90-day	180-day	365-day
Strategy return	**+309.9%**	+179.5%	+88.4%
Buy & hold return	+10.7%	+16.9%	+3.9%
Alpha	**+299.2%**	+162.6%	+84.4%
Sharpe ratio	0.90	0.75	0.61
Max drawdown	−53.0%	−56.7%	−60.1%
Buy & hold DD	−83.7%	−83.7%	−83.7%
Trades	17	11	2
Win rate	53%	73%	100%
Avg win	+25.6%	+20.4%	+37.3%
Avg loss	−5.7%	−16.5%	—

Natural gas is **the only asset where all three lookbacks produce positive alpha**: 90-day +299%, 180-day +163%, 365-day +84%. Max drawdown cut from −83.7% **to** −53–60%.

Natural gas is *violently* volatile but structurally trendless (buy & hold returned only +17% over 5 years). The momentum model turns a flat, terrifying asset into a +180–310% return by catching spikes and avoiding collapses. This is the model's ideal environment: **high volatility with no secular trend to fight**.

Part VI

Industrial Metals

Copper (HG=F): 2021–2026

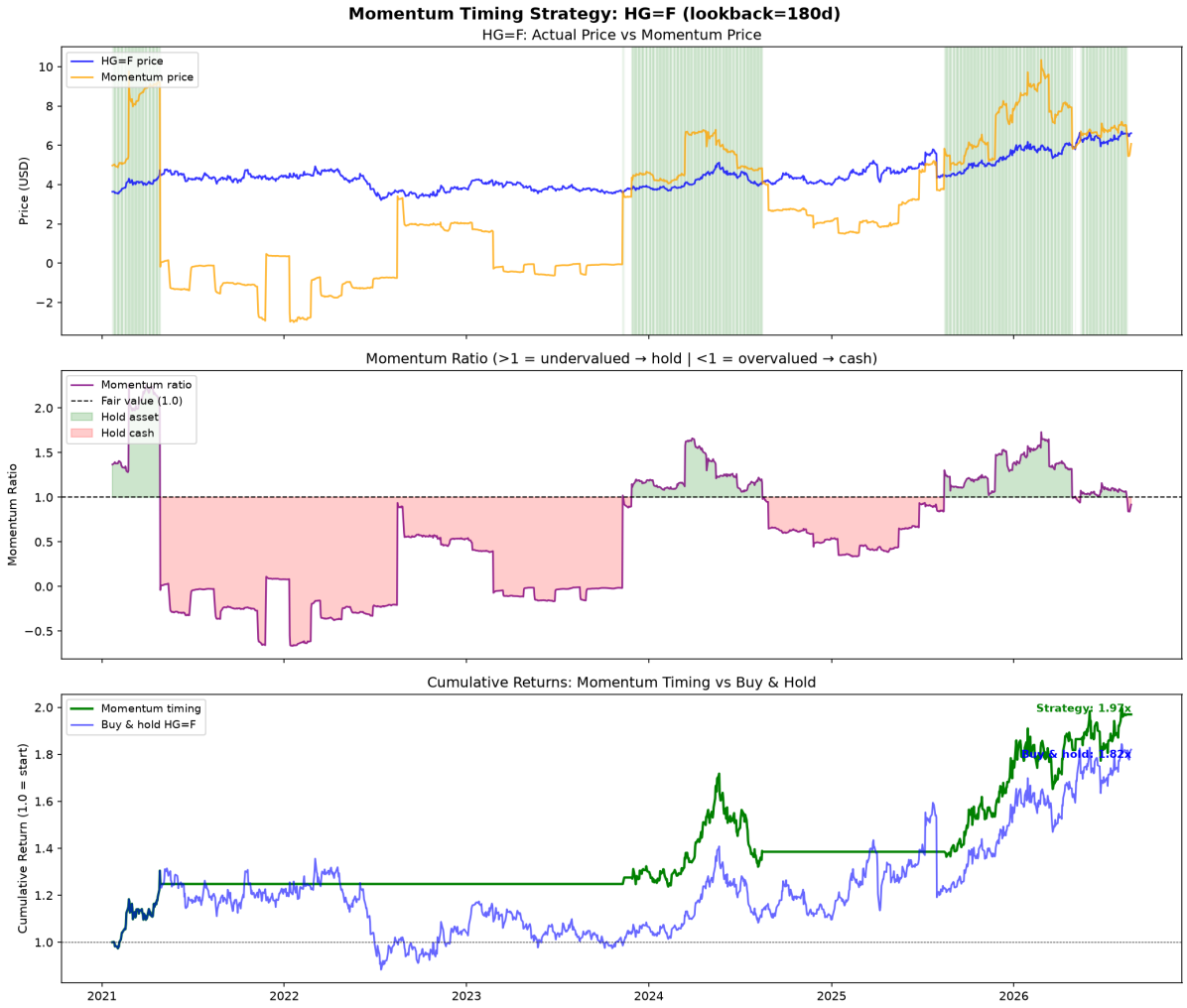


Figure 8: Copper: steady outperformance with minimal drawdown. The model catches the 2024–2025 commodity rally while avoiding the 2022 tightening-driven decline.

Results

Metric	90-day	180-day	365-day
Strategy return	+116.9%	+97.0%	+15.0%
Buy & hold return	+86.2%	+82.0%	+55.7%
Alpha	**+30.6%**	+15.0%	−40.7%
Sharpe ratio	0.96	**1.05**	0.41
Max drawdown	−25.3%	−23.1%	−11.4%
Buy & hold DD	−34.9%	−34.9%	−34.9%
Trades	10	6	4
Win rate	80%	100%	50%
Avg win	+10.5%	+12.3%	+5.8%
Avg loss	−3.4%	—	−0.7%

Copper is the **quiet achiever**: Sharpe 1.05 (third-highest after SOL and BTC), **100% win rate** (6/6 trades at 180d), max drawdown consistently cut from −35% to −11–25%. Copper occupies a middle ground: more volatile than forex, less volatile than crypto. The model extracts steady, reliable alpha without the stomach-churning drawdowns of SOL or natgas. **If you want momentum timing with the least stress, copper is the answer.**

Part VII

Equity Index

S&P 500 Futures (ES=F): 2021–2026

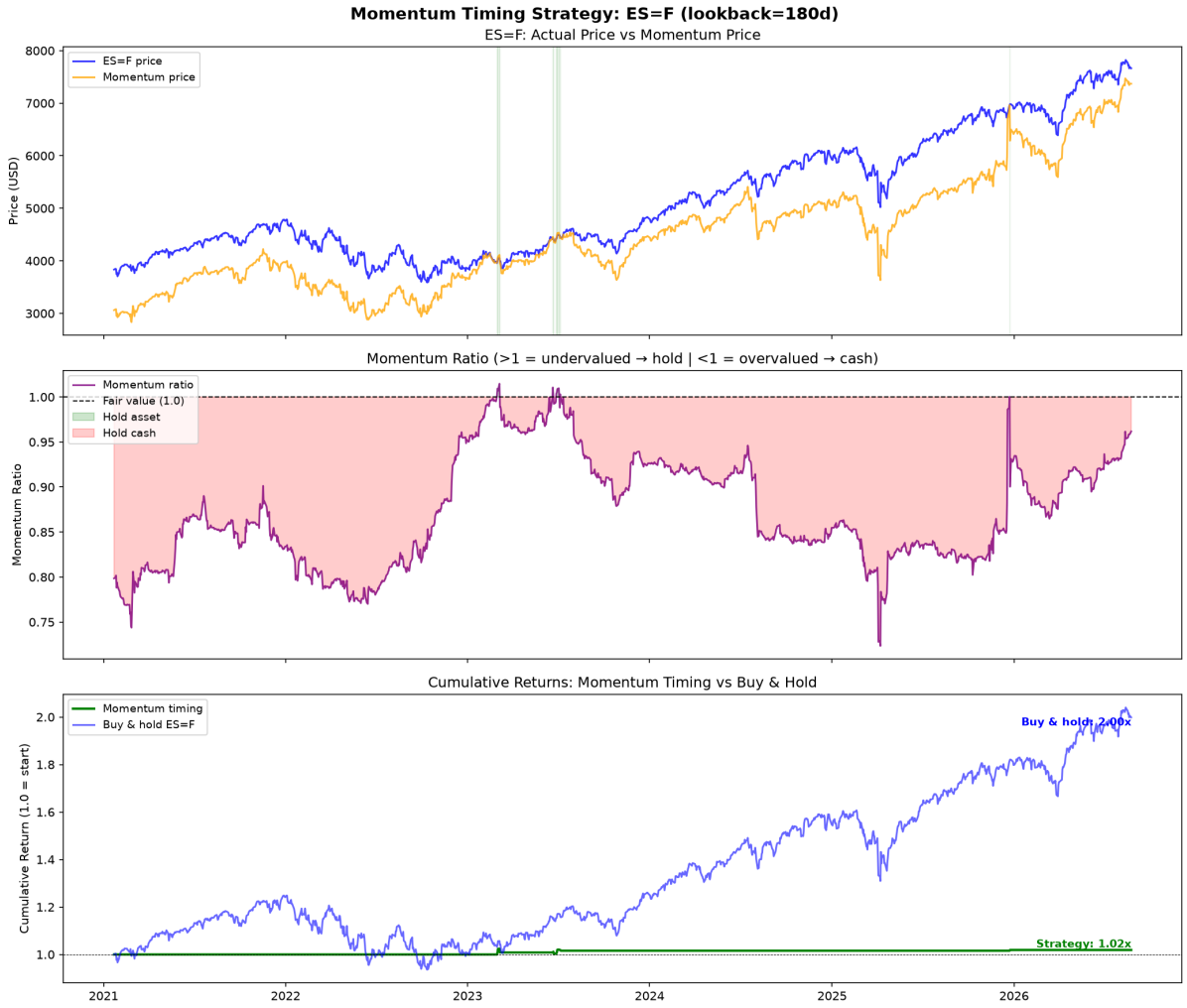


Figure 9: S&P 500 futures: the model almost never enters the market. The momentum ratio stays below 1.0 for 99% of the period, producing a flat green line while the index doubles.

Results

Metric	180-day
Strategy return	+1.9%
Buy & hold return	+99.9%
Alpha	−98.0%
Sharpe ratio	0.34
Max drawdown	−2.0%
Buy & hold DD	−25.0%
Days in asset	0.9%
Trades	5
Win rate	60%

The S&P 500 fails for a structural reason: **equity indices have a permanent upward drift** driven by earnings growth, buybacks, and index rebalancing. Volume is highest on *down* days (panic selling, margin calls), while up days have moderate volume (steady accumulation). The pool model reads this as "the big money is bearish" and keeps the momentum ratio below 1.0.

Do not use this model on equity indices.

Part VIII

Conclusions

Cross-Asset Comparison

All results using best lookback (180d unless noted). All periods 5 years (2021–2026) except gold and silver (16 years from 2010).

	SOL (180d)	BTC (180d)	NatGas (90d)	Copper (180d)	Oil (90d)	Silver (16yr)	Gold (16yr)	EUR (180d)	ES (180d)
Strategy	**+7691**	**+463**	**+310**	+97	+111	+259	+152	+7	+2
Buy&hold	+5094	+168	+11	+82	+79	+310	+334	−4	+100
Alpha	**+2597**	**+295**	**+299**	+15	+32	−51	−182	+11	−98
Sharpe	**1.34**	**1.16**	0.90	**1.05**	0.81	0.54	0.66	0.30	0.34
Max DD	−79%	−28%	−53%	−23%	−34%	−47%	−25%	−8%	−2%
B&H DD	−96%	−77%	−84%	−35%	−55%	−76%	−44%	−22%	−25%
Days in	55%	34%	50%	35%	29%	40%	36%	53%	1%
Win rate	50%	62%	53%	100%	62%	67%	60%	55%	60%

All returns in %. Best lookback chosen per asset.

What Works

1. **SOL is the strongest result in the entire study.** +2597% alpha, Sharpe 1.34, 18:1 win/loss asymmetry. Extreme volatility is the model's fuel.
2. **Natural gas has positive alpha on all lookbacks.** The only asset to achieve this. A structurally trendless, violently volatile asset is the model's ideal environment.
3. **BTC remains the flagship.** +295% alpha, Sharpe 1.16, 14:1 win/loss ratio. The most tradeable result because crypto markets are 24/7 with deep liquidity.
4. **Copper is the best risk-adjusted commodity.** Sharpe 1.05, 100% win rate (6/6 trades). Lower alpha but much lower stress.
5. **180 days is optimal for most assets.** Exception: energy commodities (oil, natgas), where 90 days captures faster-moving geopolitical and seasonal regimes.
6. **Drawdown reduction is universal.** Every single asset tested — even the failures — shows reduced max drawdown vs buy & hold. The model is better at telling you *when to get out* than when to get in.
7. **Win/loss asymmetry is universal.** Average wins are always much larger than average losses. The model cuts losers fast and lets winners run.
8. **Forex: positive alpha on all six pairs.** Best used as a risk overlay, not a return strategy.

What Doesn't Work

1. **Equity indices (S&P 500).** The structural upward drift keeps the momentum ratio permanently below 1.0. The model sits in cash 99% of the time.
2. **Pairs trading** between correlated assets (BTC/ETH). The model needs independent fundamentals.

3. **ETH timing.** Volume data is too noisy.
4. **Gold over long periods.** Extended sideways markets generate whipsaw losses.
5. **365-day lookback on most assets.** Too slow to capture regime shifts. Exception: natural gas, where it still produces +84% alpha.
6. **Forex as a return strategy.** Returns are too small to justify trading costs.

Caveats

1. **Backtest ≠ live trading.** These results assume execution at the daily close with no slippage or fees.
2. **Small samples.** 10–30 trades is not statistically robust. Strong alpha may be driven by a few large wins.
3. **Volume data quality.** Crypto exchanges inflate volume through wash trading. The signal may be noisier than it appears.
4. **No transaction costs.** Exchange fees (0.1–0.5% per trade) and bid-ask spreads will reduce returns.
5. **Tax implications.** Each trade is a taxable event. Frequent trading creates short-term capital gains at your marginal rate.
6. **Survivor bias.** We tested BTC and gold because they went up. We would not have tested assets that went to zero.
7. **Regime change.** Market structures evolve. The volume dynamics that produced these signals may not persist.

Recommendations

If you had to pick trades from all the results:

1. **Primary: SOL with 180-day momentum timing.** Highest alpha (+2597%), best Sharpe (1.34). Extreme volatility makes it high-risk but the model tames it.
2. **Core: BTC with 180-day momentum timing.** Most tradeable: deep liquidity, 24/7 markets, Sharpe 1.16.
3. **Commodity: Natural gas with 90-day timing.** +299% alpha on a structurally flat asset. Positive alpha on all lookbacks — the model's ideal environment.
4. **Conservative: Copper with 180-day timing.** Sharpe 1.05, 100% win rate, lowest stress. The grown-up trade.
5. **Risk overlay: Gold/Silver with 180-day timing.** Won't beat buy & hold, but halves your drawdown.

Next Steps

1. **Paper trade.** Run the 180-day signal in real-time for 3–6 months on BTC, SOL, copper, and natgas. No real money.
2. **Fine-tune lookback.** Test 120, 150, 200 days on BTC/SOL; test 60, 90, 120 days on energy commodities.
3. **Add transaction costs** to the simulation.

4. **Multi-asset portfolio.** Run BTC + copper + natgas simultaneously with capital allocation rules.

5. **If confirmed:** Kraken/Coinbase for crypto (CCXT library). Interactive Brokers for commodity futures (CL, NG, HG). Gold and silver via ETFs (GLD, SLV).

— *Robin*

Appendix: Source Code

All code lives in ~/git/andy/crypto/. Python 3.8+, no exchange account needed for backtesting.

Setup

Shell

```
cd ~/git/andy/crypto
python3 -m venv venv
source venv/bin/activate
pip install yfinance pandas numpy matplotlib
```

momentum_single.py — Complete File

This is the main script. It handles everything: data fetching, momentum calculation, simulation, reporting, and charting.

momentum_single.py

```
"""
Momentum timing: single-asset directional strategy.

Uses the bookmaker-pool momentum model to time entries and exits.
Hold the asset when momentum price > actual price. Go to cash
when momentum price < actual price.

All data from yfinance (free, no auth).

Usage:
  python momentum_single.py                          # BTC, defaults
  python momentum_single.py --asset ETH-USD          # ETH
  python momentum_single.py --asset GC=F             # Gold futures
  python momentum_single.py --asset SI=F             # Silver futures
  python momentum_single.py --lookback 90 --plot     # 90-day + charts
  python momentum_single.py --start 2010-01-01       # custom period
"""

import argparse
import os
import numpy as np
import pandas as pd
import yfinance as yf
from datetime import datetime, timedelta
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

SCRIPT_DIR = os.path.dirname(os.path.abspath(__file__))


# -- Core algorithm: Bookmaker Pool ---------------------------------

def compute_rolling_momentum(history, lookback_days):
    """
    Compute rolling momentum price using the bookmaker pool model.

```

```
    For each day, takes a window of `lookback_days` history, runs
    the pool model, and records the resulting momentum price and
    ratio (momentum_price / actual_price).

    Volume is normalised (divided by window mean) to prevent
    overflow on crypto and commodity volumes.
    """
    closes = history["Close"].values
    volumes = history["Volume"].values
    dates = history.index

    momentum_prices = [np.nan] * len(history)
    momentum_ratios = [np.nan] * len(history)

    for i in range(lookback_days, len(history)):
        window_close = closes[i - lookback_days:i + 1]
        window_vol = volumes[i - lookback_days:i + 1]

        if len(window_close) < 10:
            continue

        # Daily returns within the window
        returns = np.diff(window_close) / window_close[:-1]
        vols = window_vol[1:]

        # Normalise volume to prevent overflow
        mean_vol = vols.mean()
        if mean_vol == 0:
            continue
        norm_vols = vols / mean_vol

        # Net flow = normalised_volume * return
        net_flows = norm_vols * returns

        # Run the bookmaker pool
        pool = float(window_close[0])
        mp = pool

        for j in range(len(net_flows)):
            pool += net_flows[j] * mp
            mp = pool

        momentum_prices[i] = mp
        momentum_ratios[i] = mp / closes[i]

    result = pd.DataFrame({
        "Close": closes,
        "MomentumPrice": momentum_prices,
        "MomentumRatio": momentum_ratios,
    }, index=dates)

    return result.dropna()


# -- Simulation -----------------------------------------------------

def simulate_timing(asset_symbol, lookback_days,
                    start_date, end_date):
    """
    Simulate the timing strategy.

    Rule:
```

```
      - If MomentumRatio > 1.0: hold the asset
      - If MomentumRatio <= 1.0: hold cash

    Uses previous day's signal (no lookahead bias).
    """
    fetch_start = start_date - timedelta(
        days=lookback_days + 60)

    print(f"Fetching {asset_symbol}...")
    ticker = yf.Ticker(asset_symbol)
    hist = ticker.history(
        start=fetch_start, end=end_date, interval="1d")
    print(f"  Got {len(hist)} days")

    if hist.empty:
        raise ValueError(f"No data for {asset_symbol}")

    print(f"Computing rolling momentum "
          f"(lookback={lookback_days} days)...")
    df = compute_rolling_momentum(hist, lookback_days)

    # Trim to simulation period
    start_ts = pd.Timestamp(start_date)
    if df.index.tz is not None:
        start_ts = start_ts.tz_localize(df.index.tz)
    df = df[df.index >= start_ts]

    if df.empty:
        raise ValueError("No data in simulation period")

    # Signal: 1 = hold asset, 0 = hold cash
    df["signal"] = (
        df["MomentumRatio"] > 1.0).astype(int)

    # Position: use previous day's signal
    df["position"] = (
        df["signal"].shift(1).fillna(0).astype(int))

    # Returns
    df["asset_return"] = (
        df["Close"].pct_change().fillna(0))
    df["strategy_return"] = (
        df["position"] * df["asset_return"])

    # Cumulative
    df["cumulative_asset"] = (
        (1 + df["asset_return"]).cumprod())
    df["cumulative_strategy"] = (
        (1 + df["strategy_return"]).cumprod())

    return df


# -- Reporting -----------------------------------------------------

def print_report(df, asset_symbol, lookback_days):
    """Print backtest summary with trade analysis."""
    total_days = len(df)
    days_in = (df["position"] == 1).sum()
    days_out = (df["position"] == 0).sum()

    total_return = (
```

```
        df["cumulative_strategy"].iloc[-1] - 1) * 100
    asset_return = (
        df["cumulative_asset"].iloc[-1] - 1) * 100

    transitions = (
        df["position"].diff().abs() > 0).sum()

    # Sharpe (annualised, 365 days)
    sr = df["strategy_return"]
    sharpe = (sr.mean() / sr.std() * np.sqrt(365)
              if sr.std() > 0 else 0)

    # Max drawdown
    cummax = df["cumulative_strategy"].cummax()
    drawdown = (
        df["cumulative_strategy"] - cummax) / cummax
    max_dd = drawdown.min() * 100

    bh_cummax = df["cumulative_asset"].cummax()
    bh_drawdown = (
        df["cumulative_asset"] - bh_cummax) / bh_cummax
    bh_max_dd = bh_drawdown.min() * 100

    # Trade-level analysis
    in_blocks = []
    block_start = None
    for i in range(len(df)):
        if (df["position"].iloc[i] == 1
                and block_start is None):
            block_start = i
        elif (df["position"].iloc[i] == 0
                and block_start is not None):
            ret = (df["Close"].iloc[i]
                   / df["Close"].iloc[block_start]
                   - 1) * 100
            in_blocks.append(ret)
            block_start = None
    if block_start is not None:
        ret = (df["Close"].iloc[-1]
               / df["Close"].iloc[block_start]
               - 1) * 100
        in_blocks.append(ret)

    wins = sum(1 for r in in_blocks if r > 0)
    losses = sum(1 for r in in_blocks if r <= 0)
    avg_win = (np.mean([r for r in in_blocks if r > 0])
               if wins > 0 else 0)
    avg_loss = (np.mean([r for r in in_blocks if r <= 0])
                if losses > 0 else 0)

    # Crashes avoided
    out_blocks = []
    block_start = None
    for i in range(len(df)):
        if (df["position"].iloc[i] == 0
                and block_start is None):
            block_start = i
        elif (df["position"].iloc[i] == 1
                and block_start is not None):
            avoided = (df["Close"].iloc[i]
                       / df["Close"].iloc[block_start]
                       - 1) * 100
```

```
            out_blocks.append(avoided)
            block_start = None
    if block_start is not None:
        avoided = (df["Close"].iloc[-1]
                   / df["Close"].iloc[block_start]
                   - 1) * 100
        out_blocks.append(avoided)

    crashes_avoided = sum(
        1 for r in out_blocks if r < -10)

    sym = asset_symbol[:3]
    print()
    print("=" * 60)
    print(f"MOMENTUM TIMING: {asset_symbol}")
    print("=" * 60)
    print(f"  Period:  "
          f"{df.index[0].strftime('%Y-%m-%d')} -> "
          f"{df.index[-1].strftime('%Y-%m-%d')}")
    print(f"  Lookback: {lookback_days} days")
    print(f"  Days in {sym}: {days_in} "
          f"({days_in/total_days*100:.1f}%)")
    print(f"  Days cash:  {days_out} "
          f"({days_out/total_days*100:.1f}%)")
    print(f"  Trades: {len(in_blocks)}")
    print()
    print(f"  Strategy:  {total_return:+.1f}%")
    print(f"  Buy&Hold:  {asset_return:+.1f}%")
    print(f"  Alpha:     "
          f"{total_return - asset_return:+.1f}%")
    print(f"  Sharpe:    {sharpe:.2f}")
    print(f"  Max DD:    {max_dd:.1f}% "
          f"(hold: {bh_max_dd:.1f}%)")
    if in_blocks:
        print(f"  Win rate:  "
              f"{wins}/{len(in_blocks)} "
              f"({wins/len(in_blocks)*100:.0f}%)")
        print(f"  Avg win:   {avg_win:+.1f}%  "
              f"Avg loss: {avg_loss:+.1f}%")
    print(f"  Crashes avoided: {crashes_avoided}")
    print()

    return {
        "total_return": total_return,
        "asset_return": asset_return,
        "sharpe": sharpe,
        "max_dd": max_dd,
        "bh_max_dd": bh_max_dd,
        "trades": len(in_blocks),
        "wins": wins, "losses": losses,
        "avg_win": avg_win, "avg_loss": avg_loss,
    }


# -- Charting ------------------------------------------------

def plot_results(df, asset_symbol, lookback_days,
                 stats, tag=""):
    """Generate 3-panel chart."""
    fig, axes = plt.subplots(
        3, 1, figsize=(14, 12), sharex=True)

```

```
    # Panel 1: Price vs momentum price
    ax = axes[0]
    ax.plot(df.index, df["Close"],
            label=f"{asset_symbol} price",
            color="blue", alpha=0.8)
    ax.plot(df.index, df["MomentumPrice"],
            label="Momentum price",
            color="orange", alpha=0.8)
    for i in range(1, len(df)):
        if df["position"].iloc[i] == 1:
            ax.axvspan(df.index[i-1], df.index[i],
                       alpha=0.05, color="green")
    ax.set_ylabel("Price (USD)")
    ax.set_title(f"{asset_symbol}: "
                 f"Actual Price vs Momentum Price")
    ax.legend(loc="upper left", fontsize=9)

    # Panel 2: Momentum ratio
    ax = axes[1]
    ax.plot(df.index, df["MomentumRatio"],
            label="Momentum ratio",
            color="purple", alpha=0.8)
    ax.axhline(y=1.0, color="black", linewidth=1,
               linestyle="--", label="Fair value (1.0)")
    ax.fill_between(
        df.index, df["MomentumRatio"], 1.0,
        where=df["MomentumRatio"] > 1.0,
        alpha=0.2, color="green", label="Hold asset")
    ax.fill_between(
        df.index, df["MomentumRatio"], 1.0,
        where=df["MomentumRatio"] < 1.0,
        alpha=0.2, color="red", label="Hold cash")
    ax.set_ylabel("Momentum Ratio")
    ax.set_title("Momentum Ratio "
                 "(>1 = hold | <1 = cash)")
    ax.legend(loc="upper left", fontsize=9)

    # Panel 3: Cumulative returns
    ax = axes[2]
    ax.plot(df.index, df["cumulative_strategy"],
            label="Momentum timing",
            color="green", linewidth=2)
    ax.plot(df.index, df["cumulative_asset"],
            label=f"Buy & hold {asset_symbol}",
            color="blue", alpha=0.6)
    ax.axhline(y=1.0, color="black",
               linewidth=0.5, linestyle="--")

    sf = df["cumulative_strategy"].iloc[-1]
    hf = df["cumulative_asset"].iloc[-1]
    ax.annotate(f"Strategy: {sf:.2f}x",
                xy=(df.index[-1], sf),
                fontsize=9, fontweight="bold",
                color="green", ha="right", va="bottom")
    ax.annotate(f"Buy & hold: {hf:.2f}x",
                xy=(df.index[-1], hf),
                fontsize=9, fontweight="bold",
                color="blue", ha="right", va="top")

    ax.set_ylabel("Cumulative Return")
    ax.set_title("Cumulative Returns: "
                 "Timing vs Buy & Hold")
```

```
    ax.legend(loc="upper left", fontsize=9)

    fig.suptitle(
        f"Momentum Timing: {asset_symbol} "
        f"(lookback={lookback_days}d)",
        fontsize=14, fontweight="bold")
    fig.tight_layout()

    out = os.path.join(
        SCRIPT_DIR,
        f"timing_{asset_symbol.replace('-','')}"
        f"{tag}.png")
    fig.savefig(out, dpi=150)
    plt.close(fig)
    print(f"Saved: {out}")
    return out


# -- CLI ------------------------------------------------------------

def main():
    parser = argparse.ArgumentParser(
        description="Momentum timing backtest")
    parser.add_argument(
        "--asset", default="BTC-USD",
        help="Ticker symbol. Examples: BTC-USD, "
             "ETH-USD, GC=F (gold), SI=F (silver)")
    parser.add_argument(
        "--lookback", type=int, default=180,
        help="Lookback window in days (default: 180)")
    parser.add_argument(
        "--start", default=None,
        help="Start date YYYY-MM-DD (default: 5yr ago)")
    parser.add_argument(
        "--end", default=None,
        help="End date YYYY-MM-DD (default: today)")
    parser.add_argument(
        "--plot", action="store_true",
        help="Generate PNG charts")

    args = parser.parse_args()

    end_date = (
        datetime.strptime(args.end, "%Y-%m-%d")
        if args.end else datetime.now())
    start_date = (
        datetime.strptime(args.start, "%Y-%m-%d")
        if args.start else
        end_date - timedelta(days=5*365))

    print(f"\n  Asset:    {args.asset}")
    print(f"  Lookback: {args.lookback} days")
    print(f"  Period:   "
          f"{start_date.strftime('%Y-%m-%d')} -> "
          f"{end_date.strftime('%Y-%m-%d')}\n")

    df = simulate_timing(
        args.asset, args.lookback,
        start_date, end_date)

    stats = print_report(
        df, args.asset, args.lookback)
```

```
410
411     if args.plot:
412         plot_results(
413             df, args.asset, args.lookback, stats)
414
415     csv_out = os.path.join(
416         SCRIPT_DIR,
417         f"timing_{args.asset.replace('-','')}.csv")
418     df.to_csv(csv_out)
419     print(f"Saved: {csv_out}")
420
421
422 if __name__ == "__main__":
423     main()
```

Running the Experiments

Reproduce all results from this report

```
# Bitcoin (5 years, 3 lookback windows)
python momentum_single.py --asset BTC-USD --lookback 90 \
       --start 2021-01-01 --plot
python momentum_single.py --asset BTC-USD --lookback 180 \
       --start 2021-01-01 --plot
python momentum_single.py --asset BTC-USD --lookback 365 \
       --start 2021-01-01 --plot

# Ethereum
python momentum_single.py --asset ETH-USD --lookback 180 \
       --start 2021-01-01 --plot

# Solana (5 years, 3 lookback windows)
python momentum_single.py --asset SOL-USD --lookback 90 \
       --start 2021-01-01 --plot
python momentum_single.py --asset SOL-USD --lookback 180 \
       --start 2021-01-01 --plot
python momentum_single.py --asset SOL-USD --lookback 365 \
       --start 2021-01-01 --plot

# Gold (16 years)
python momentum_single.py --asset GC=F --lookback 90 \
       --start 2010-01-01 --plot
python momentum_single.py --asset GC=F --lookback 180 \
       --start 2010-01-01 --plot
python momentum_single.py --asset GC=F --lookback 365 \
       --start 2010-01-01 --plot

# Silver (16 years)
python momentum_single.py --asset SI=F --lookback 90 \
       --start 2010-01-01 --plot
python momentum_single.py --asset SI=F --lookback 180 \
       --start 2010-01-01 --plot
python momentum_single.py --asset SI=F --lookback 365 \
       --start 2010-01-01 --plot

# Recent period only (gold & silver)
python momentum_single.py --asset GC=F --lookback 180 \
       --start 2021-01-01 --plot
python momentum_single.py --asset SI=F --lookback 180 \
       --start 2021-01-01 --plot
```

```
# Crude Oil (5 years, 3 lookback windows)
python momentum_single.py --asset CL=F --lookback 90 \
       --start 2021-01-01 --plot
python momentum_single.py --asset CL=F --lookback 180 \
       --start 2021-01-01 --plot
python momentum_single.py --asset CL=F --lookback 365 \
       --start 2021-01-01 --plot

# Natural Gas (5 years, 3 lookback windows)
python momentum_single.py --asset NG=F --lookback 90 \
       --start 2021-01-01 --plot
python momentum_single.py --asset NG=F --lookback 180 \
       --start 2021-01-01 --plot
python momentum_single.py --asset NG=F --lookback 365 \
       --start 2021-01-01 --plot

# Copper (5 years, 3 lookback windows)
python momentum_single.py --asset HG=F --lookback 90 \
       --start 2021-01-01 --plot
python momentum_single.py --asset HG=F --lookback 180 \
       --start 2021-01-01 --plot
python momentum_single.py --asset HG=F --lookback 365 \
       --start 2021-01-01 --plot

# S&P 500 Futures (the failure case)
python momentum_single.py --asset ES=F --lookback 180 \
       --start 2021-01-01 --plot

# Forex futures (CME, 5 years)
python momentum_single.py --asset 6E=F --lookback 180 \
       --start 2021-01-01 --plot    # EUR
python momentum_single.py --asset 6A=F --lookback 180 \
       --start 2021-01-01 --plot    # AUD
python momentum_single.py --asset 6B=F --lookback 180 \
       --start 2021-01-01 --plot    # GBP
python momentum_single.py --asset 6J=F --lookback 180 \
       --start 2021-01-01 --plot    # JPY
python momentum_single.py --asset 6C=F --lookback 180 \
       --start 2021-01-01 --plot    # CAD
python momentum_single.py --asset 6S=F --lookback 180 \
       --start 2021-01-01 --plot    # CHF

# Try any yfinance ticker:
python momentum_single.py --asset AAPL --lookback 180 --plot
python momentum_single.py --asset BHP.AX --lookback 180 --plot
```